

Smart Health Symptom Checker & Disease Prediction System

Overview

The Smart Health Symptom Checker is a simple, user-friendly web application that helps people understand possible diseases based on the symptoms they enter. It uses machine learning to suggest potential conditions, offers basic precautionary tips, and can generate a simple health report. The goal of the project is to make basic medical knowledge more accessible.

Student Details

Name: Sanjana Yadav

Reg. No.: 25BCY10270

Branch: CSE

College: VIT Bhopal

Key Features

- Easy symptom input with a clean interface
- Machine-learning based disease prediction
- Prevention tips and basic suggestions
- Optional health report generation
- Optional admin module for dataset management

Non-Functional Highlights

- Fast response time (under 3 seconds)
- Secure login and password protection
- Scalable for many users
- Clean and simple UI
- Reliable system with backups
- Proper validation and error handling

Tech Stack

Frontend: HTML, CSS, JavaScript, React

Backend: Python (Flask/Django) or Node.js

Database: MongoDB or MySQL

ML Models: Naive Bayes, Random Forest, SVM

Hosting: GitHub, Render, Netlify

System Diagrams

Includes: Use case diagram, workflow diagram, system architecture, ER diagram

Future Possibilities

- Voice-based symptom entry
- Live doctor consultation
- Integration with smart wearable devices